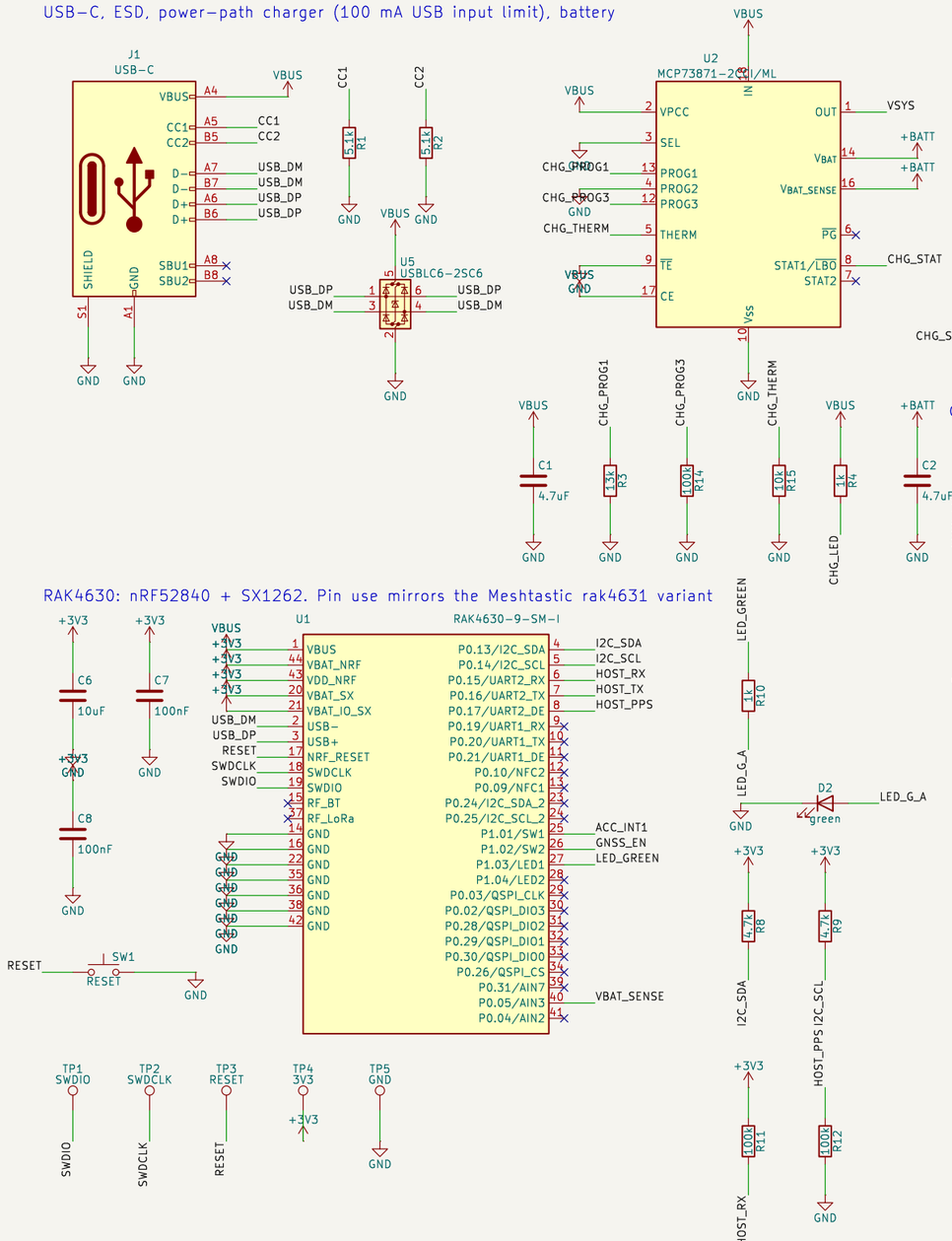
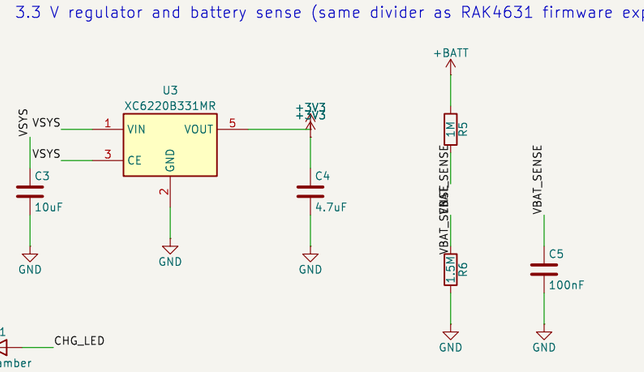
Willie Finder tracker — draft schematic rev 0.3 (not for fabrication)

USB-C, ESD, power-path charger (100 mA USB input limit), battery



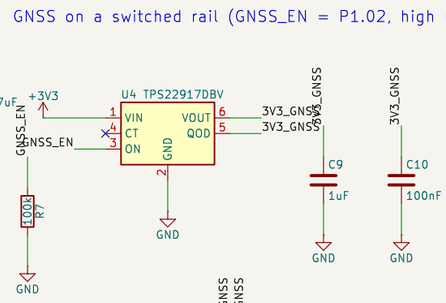
3.3 V regulator and battery sense (same divider as RAK4631 firmware expects)

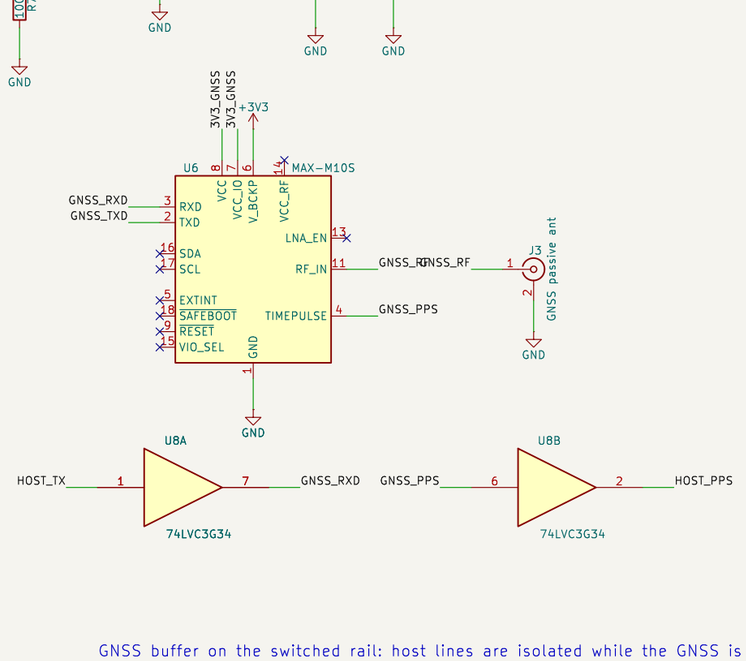


U2: SEL and PROG2 low = USB, 100 mA input limit. The cell supplements the load above that. Only C1 (4.7 uF) sits directly on VBUS.

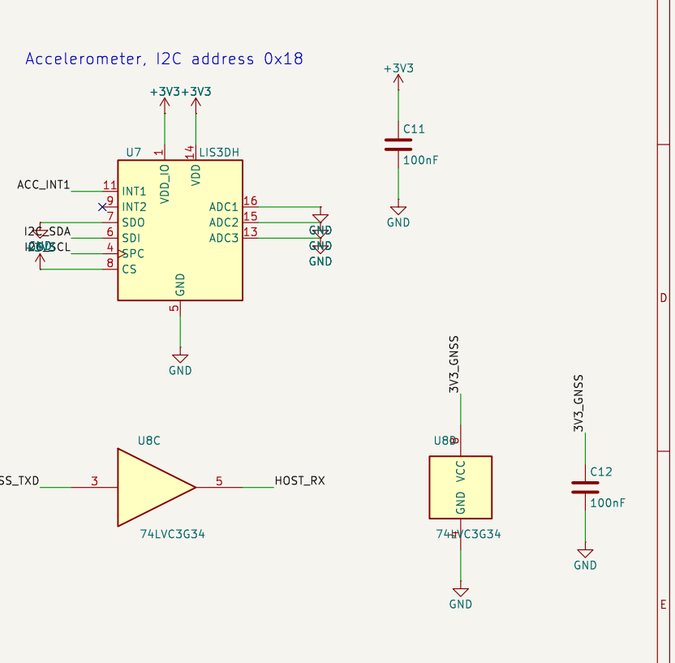
RAK4630: nRF52840 + SX1262. Pin use mirrors the Meshtastic rak4631 variant

GNSS on a switched rail (GNSS_EN = P1.02, high = on)





Accelerometer, I2C address 0x18



GNSS buffer on the switched rail: host lines are isolated while the GNSS is off

LoRa and Bluetooth antennas plug into the U.FL sockets on the RAK4630 itself (-I variant).

J3 is for a passive GNSS antenna only: there is no DC bias on the coax. Route GNSS_RF as 50 ohm.

MAX-M10S VIO_SEL left open = 3.3 V I/O. Battery must have its own protection circuit.

Battery reading: divider is nominally x1.667, stock firmware uses x1.73. Calibrate against a meter.

Stacked pins (several pads share one drawn pin; all are in the netlist):

U6 GND = 1, 10, 12. U7 GND = 5, 10, 12. U2 IN = 18, 19; OUT = 1, 20; VBAT = 14, 15; VSS = 10, 11, EP(21). J1 VBUS = A4, A9, B4, B9; GND = A1, A12, B1, B12.

Rev 0.3: power-path charger replaces discrete load sharing. Generated from pcb/gen_schematic.py.

Sheet: /
File: willie_tracker.kicad_sch

Title: Willie Finder tracker

Size: A3	Date:	**Rev: 0.3 draft**
KiCad E.D.A. 9.0.9		Id: 1/1